Commensurability of Observation Protocols: When Coherent Windows Determine a Process

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Abstract

Observing a process through overlapping finite windows yields local statistics; we ask when locally coherent window-statistics must arise from a single global process. This is a *skeleton* toward that target: the architecture and its status, not a finished paper. An observation protocol is *commensurable* when empirical adequacy (pairwise coherence, EA) forces realisability by a declared class (PR($\mathcal{R}$)); the failure is contextuality in the sense of Abramsky–Brandenburger, taken from the observational side. Geometrically, a protocol is commensurable iff its coherence polytope C equals its realisability polytope R, so contextuality is an exactly computable separation with a rational witness. At a fixed length the boundary is classical: for a symbolic language observed around a ring, a length is contextual exactly when the layered ring carries a layer-injective winding-≥ 2 cycle, which is the classification of contextual vertices of cycle scenarios (Kharoof–Ipek–Okay 2024; Barrett et al. 2005 and Araújo et al. 2013 for special cases). Across all lengths of one language the safe set is eventually periodic and computable, which is elementary given that classification (Lean-verified). The reconstruction guarantee holds unconditionally on three tame families and would hold for all total languages if the catalogue of taming mechanisms were complete — the open *universal impossibility* capstone. We reduce that capstone to a single named forcing lemma (primitivity forces a parity obstruction), and that lemma to a reach lemma about two synchronous tokens on the language's graph, validated without counterexample but not proved. That lemma is what remains, and it is the paper's only claimed open contribution. The worked results that populate the criterion — the parity theorem and its PR-box, the taming catalogue, Circuit Localization — are established and deferred to the body; the skeleton points at where they slot beneath the target.

1 The target: when do coherent windows determine a process?

A process in time is often seen only through *windows*: overlapping finite views, each recording finitely many coordinates over finitely many steps. Each window yields a local probability table. The tables may agree wherever they overlap and still be produced by no single underlying process. The question of this paper is when that cannot happen.

Goal 1.1 (Reconstruction guarantee — the target)**.** For a structured class of observation protocols, locally coherent window-statistics are induced by a single global process; and the boundary of that class — where coherence stops forcing a global process — is exactly decidable.

Status of the target. The guarantee is established **unconditionally** on three tame families (§4), and **conditionally** — on a single open conjecture (§5) — for all total symbolic languages. At a fixed observation length the boundary is **[CLASSICAL]**: it is the classification of contextual vertices of cycle scenarios (§4). The lift to all lengths of one language — Safe(ρ) eventually periodic and effectively computable — is **[PROVED, Lean-verified]** but elementary given that classification. The paper's own open content is the capstone of §5.

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The rest of the paper is the dependency chain beneath Goal 1.1: the geometric reduction that makes "coherence forces a process" computable (§2), why it cannot be settled at the level of individual models (§3), the criterion that draws the boundary (§4), and the conjecture that would make the guarantee unconditional (§5).

2 The geometric reduction

We fix the minimal vocabulary once, one object per sentence, and use nothing before it is defined.

A finite *alphabet* A and a finite index set $F \subseteq \mathbb{Z}$ give a *context*: a partition of A^F into cells. A *protocol* is a finite family of contexts. *Window-data* assigns a probability to each cell of each context. A *symbolic language* is a relation $\rho \subseteq A \times A$ listing the legal adjacent pairs of a shift of finite type; a protocol *reports* on the configurations legal for ρ.

Definition 2.1 (coherence, realisability, commensurability)**.** Window-data is *empirically adequate* (EA) if any two contexts agree on the events common to both. It is *realisable in a class* $\mathcal{R}$ (PR($\mathcal{R}$)) if some member of $\mathcal{R}$ — all measures, or stationary, or subshift-supported, or memory-m Markov: an observer's declared commitment — induces it. Data is *contextual* when EA but not PR($\mathcal{R}$). A protocol is *commensurable* when every EA datum is PR($\mathcal{R}$).

Remark 2.2 (Paper-local notation)**.** Here EA is overlap compatibility of local probability tables and PR($\mathcal{R}$) is realisability by one process in a declared class. The companion paper on distributivity and realism uses EA for a dual-space measure and PR for descent to an underlying realisation space. No translation between those source-specific definitions is currently established.

Proposition 2.3 (commensurability is a polytope equality)**.** *The* EA *data form a polytope C (the coherence polytope) and the* PR($\mathcal{R}$) *data a polytope $R \subseteq C$ (the realisability polytope, the convex hull of the global configurations' cell-incidence vectors). A protocol is commensurable iff $C = R$; a contextual datum is a point of $C \setminus R$, and over rational data a separating facet certifies it.*

[PROVED, CLASSICAL polyhedral content.] This is the marginal polytope of the Abramsky–Brandenburger presheaf model. Its force here is methodological: "coherence forces a process" becomes the exactly computable question $C = R$, decided by a proof of equality or an exhibited rational witness in $C \setminus R$.

3 Why the question lives at the protocol level

The reduction $C = R$ is computable per protocol but not classifiable per model: the achievable geometry is already universal.

Theorem 3.1 (trichotomy)**.** *Reporting splits into three regimes.* (Acyclic) *If the reporting hypergraph is acyclic, the protocol is commensurable for every ρ.* (Product) *Full-support coordinate protocols are commensurable.* (Variety-typed) *Protocols carrying a constraint variety realise, as their coherence polytopes, every rational polytope up to affine equivalence.*

[PROVED; the three parts are CLASSICAL translations] — acyclic = Vorob'ev/running-intersection; product = product measure; variety-typed = De Loera–Onn universality with the Pitowsky NP-hardness floor. *Consequence, and the reason for the paper's altitude:* no finite facet taxonomy classifies contextual *models*, so the classification is posed one level up — a $\forall\exists$ question over structured protocol *families* — where structure can defeat the universality. Everything below characterises such families.

4 The boundary: the winding criterion

[Repositioned 2026-10-15.] At fixed L this section is classical, and is cited rather than claimed. The contextual vertices of the n-cycle scenario with outcomes $\mathbb{Z}_d$ are exactly the k-order cycle distributions, $k \geq 2$ [4, Thm. 4.4]; the cases $n = 4$ [3] and binary outcomes [2, Thm. 2] came first, and in any scenario the vertices split into deterministic and minimal-support strongly contextual models [1, Prop. 6.3]. A k-order cycle distribution is exactly a layer-injective simple cycle of winding k below.

Read a cyclic protocol as recurrence: observe ρ around a ring of length L. Lift the ring to its *layered graph* — vertices (i, a) for site $i \in \mathbb{Z}_L$ and state $a \in A$, arcs $(i, a) \to (i{+}1, b)$ for legal pairs (a, b). The *winding* of a closed walk is its net site-advance divided by L. Call a length L *unsafe* for ρ if the length-L ring carries contextual data.

Theorem 4.1 (winding criterion; a corollary of [4, Thm. 4.4])**.** *A length L is unsafe for ρ iff the layered ring carries a layer-injective simple cycle of winding ≥ 2.*

[**CLASSICAL at fixed L**]: restrict [4, Thm. 4.4] to the face of data supported on ρ, whose vertices are vertices of the whole polytope. An independent proof, reached before the citation was found, uses three obligations, each discharged in the archive: a *gate identification* (C is the layered graph's circulation polytope) [**PROVED; the genericity gate is vacuous for the ring used**]; a *flow decomposition* (a polytope vertex is a single simple cycle) [**PROVED on a CLASSICAL, axiomatised decomposition**]; and a *layer-injectivity discriminant* (that cycle is layer-injective) [**PROVED, machine-verified in Lean 4**]. [**SCOPE: the criterion as proved here is about winding-2 obstructions; that this captures full safety rests on the pruning lemma, §5. As of 2026-08-21 that lemma is PROVED at general k and machine-verified in Lean 4 (`QuerySystem/PruningTheorem.lean`, `TheoremB.lean`), so the bridge is no longer spot-checked. It does not close the capstone: see the forcing lemma.**]

Instances of the criterion (deferred; proved in the archive). The criterion is exercised, not merely stated, on worked families. The *parity theorem* — the golden-mean shift on a ring is commensurable iff L is even, the odd-ring gap a single extreme point that is its own PR-box — is the smallest nontrivial instance. A *catalogue of ten taming mechanisms* gives sufficient conditions for the winding obstruction never to form, organised as terminal integrality certificates versus reductions to one. Assembled, these yield *Circuit Localization* for symmetric loopless marginal-determined languages, where contextuality is excluded by structure rather than search. These are [**PROVED**] and belong in the full paper; here they are the evidence that the boundary is real and computable, not the target.

5 The capstone: universal impossibility

The winding criterion decides each length. The catalogue of §4 gives sufficient conditions for a language to be safe at *all* lengths. Whether those conditions are also *necessary* — whether the catalogue is complete — is the open capstone that would make Goal 1.1 unconditional.

Conjecture 5.1 (universal impossibility)**.** No *fork* exists: every total language is either caught by a catalogued taming or is contextual at cofinitely many lengths. Equivalently, the winding criterion is a complete decision procedure and the reconstruction guarantee is unconditional on the total languages.

[**OPEN**]**.** The symmetric case is settled by Circuit Localization; the residue is the strongly-connected non-symmetric class, where exhaustive search finds no fork at small scale. The intended route reduces the conjecture to a single named lemma.

Named Lemma 5.2 (the forcing lemma — the one open step)**.** On a primitive language, an infinite safe set forces a parity obstruction: the signed pair-graph of the language is balanced (equivalently, its residue class is the even lengths), the only alternative being a finite safe set.

[OPEN; VALIDATED, not proved]. The forcing lemma is exactly "the tamings are the only sources of residue-safety in a primitive language." Its two halves are: a *re-characterisation* (residue-safety = balance of a signed pair-graph) **[PROVED, structural]**; and the *forcing* proper (primitivity excludes the mixed cases), framed as two Perron–Frobenius exclusions. The crossed-cycle exclusion is **[PROVED by hand, machine-checked stepwise]**. The aperiodicity exclusion is reduced as follows. In the pair graph (two tokens moving synchronously, never on the same state), the chase configurations on all simple cycles form a single aperiodic class **[PROVED by hand]**. Reversing arcs, membership in that class follows from forward reachability. So the forcing lemma follows from one statement.

Named Lemma 5.3 (reach lemma — the open core)**.** On a primitive language, from every recurrent pair of distinct states the two tokens can reach, without colliding, a pair lying on a common simple cycle.

[OPEN; VALIDATED, not proved]. - Primitivity is essential: imprimitive languages give periodic counterexamples to the forcing lemma's conclusion. - An explicit strategy reaches such a pair from every recurrent pair, with no failures exhaustively up to four states and on samples up to seven. In the strategy, one token circles a cycle and the other lands on it out of phase. - Its proof is complete except for one case: every off-cycle region drains into the cycle.

Discharging it closes Conjecture 5.1: the other half, the winding-2-to-full bridge, is no longer outstanding — the pruning theorem is proved at general k and Lean-verified, and with the k-cap gives $\mathrm{Unsafe}(\rho) = \bigcup_{k=2}^{|A|} \mathrm{TR}_k^{(1)}$, so every winding is accounted for. **The reach lemma is now the single remaining result the paper is aiming to prove or to state as its open problem.**

6 Consequences

On the tame classes of §4 — acyclic reporting, symmetric marginal-determined languages, even-cyclic golden-mean recurrence — coherent window-statistics determine a global process (Goal 1.1, unconditional), and the winding criterion makes the boundary decidable. This is the licensing layer beneath process reconstruction in the statistical (measure-plus-shift) rather than manifold sense of Takens. By the universality of §3 the boundary is intrinsic, so contextuality is a feature of observing through windows: present exactly when the geometry admits the winding obstruction. Where Conjecture 5.1 holds, no total language manufactures contextuality unless the exchange criterion fires.

The same failure of local coherence to globalise recurs across the programme: algebraically in Paper II (dispersion-free states collapsing realisability on non-distributive logics) and set-theoretically in the σ-essential witness (a finitely coherent pattern with no σ-additive global state). Dynamically it is sharpest — an unwitnessed exchange, two histories trading places around the clock of observation without meeting.

Positioning and method

Contribution type. A bridge (Type 6): the polyhedral core is classical (marginal polytopes, stable-set integrality, De Loera–Onn universality, signed-minor tightness); the novelty is the observational-dynamical framing, the winding criterion, and the Circuit-Localization assembly. This positioning is stated, not hedged.

Method. Each commensurability verdict is a theorem or an exact rational enumeration; each contextuality verdict an exhibited rational witness in $C \setminus R$; sampling is only screening. The layer-injectivity discriminant and the flow-decomposition kernel are machine-checked in Lean 4.

What is owed to fill this skeleton to a complete paper. (i) The forcing lemma (Lemma 5.2), now reduced to the reach lemma (Lemma 5.3): prove it, or state it as the paper's open problem. (ii) The winding-2-to-full-safety bridge: prove it applies, or scope every boundary claim to winding-2. (iii) Fold the archived proved body (parity, PR-box, the ten tamings, Circuit Localization, the three winding obligations) back in as §4's instances, ordered define-at-first-use.

References

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